
THE LITTLEFISH HUNTER

Enterprise Cross-Environment Threat Intelligence Briefing

CYBER THREAT INTELLIGENCE MONTHLY BRIEFING

Operational Window: Last 30 Days (June - July 2026)

Document Classification: TLP:CLEAR

Target Environments: Commercial Data Center Architecture & Residential Edge Decoys

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1. Executive Summary & Core Objective

Automated perimeter probing remains the primary mechanism for opportunistic threat groups seeking to identify exploitable vectors across public network ranges. This enterprise intelligence briefing documents a comprehensive 30-day monitoring exercise executed under the security project framework '**The Littlefish Hunter**'. Telemetry aggregation was driven by a centralized Wazuh SIEM deployment, ingesting and parsing more than 250,000 discrete authentication actions.

The focal point of this research is a comparative and behavioral analysis of internet-wide scanning operations across two completely segregated environments: an enterprise cloud infrastructure private instance and a residential edge decoy appliance. The telemetry obtained reveals an asymmetric, highly specialized mapping of botnet assets, indicating that modern threat architectures segment their target pools by infrastructure category, deploying completely unique dictionaries and scanning methods depending on the network classification of the target host.

2. Scope & Boundaries

The tactical parameters of this operation were tightly bounded to guarantee data integrity and eliminate cross-contamination from local network testing.

Parameter	Defined Operational Scope
Temporal Window	30 Continuous Days (June 02, 2026 – July 02, 2026)
Protocol Exposure	TCP Port 22 (Secure Shell - SSH / Simulated Telnet)
Inbound Ingestion Volume	254,810 Logged Events via Wazuh Host Engines

3. Methodology Framework

The analytical lifecycle adheres strictly to standard threat processing paradigms, converting unstructured event data into actionable cyber threat intelligence. Raw events generated at the operating system kernel and application layers were monitored via native log rotation engines. The localized Wazuh agent daemons extracted these log rows, packaged them into structured JSON payloads, and pushed them across encrypted TLS tunnels to the centralized manager architecture.

4. Environment Overview

- **Environment Alpha (Centralized Management SIEM Core):** Hardened Debian Linux host executing the Wazuh SIEM Manager architecture.
- **Environment Beta (Production Cloud Boundary Assets):** Ubuntu 24.04.4 LTS server instance exposed on a public data center IP (Agent PL84924).
- **Environment Gamma (Residential IoT Network Decoy):** Raspberry Pi lab appliance running an interactive, high-fidelity Cowrie Honeypot script (Agent rpilab).

5. Data Collection & Transport Logistics

Log files on monitored nodes are processed via internal pipeline queues. For Environment Beta, standard Linux PAM and sshd subsystems logs are funneled to `/var/log/auth.log`. For Environment Gamma, Cowrie intercepts connection states, outputting full event records into a structured local `cowrie.json` file.

6. Attack Surface Mapping

The exposure surfaces of both target nodes were intentionally left visible to global network scans to establish a clear baseline. For Environment Beta (Cloud VPS), the SSH service remained on standard port 22. However, password authentication for administrative access was disabled at the service layer, making it an un-exploitable wall.

For Environment Gamma (Honeypot), Port 22 was configured to simulate a vulnerable OpenSSH shell on an unpatched Linux embedded device, intentionally granting access to standard brute-force dictionaries.

7. Global Threat Landscape Context

The current global threat landscape is saturated with automated scanning engines running on compromised servers and cheap bulletproof hosting providers. These platforms are managed by botnets like Mirai, Mozi, or specialized brute-force networks.

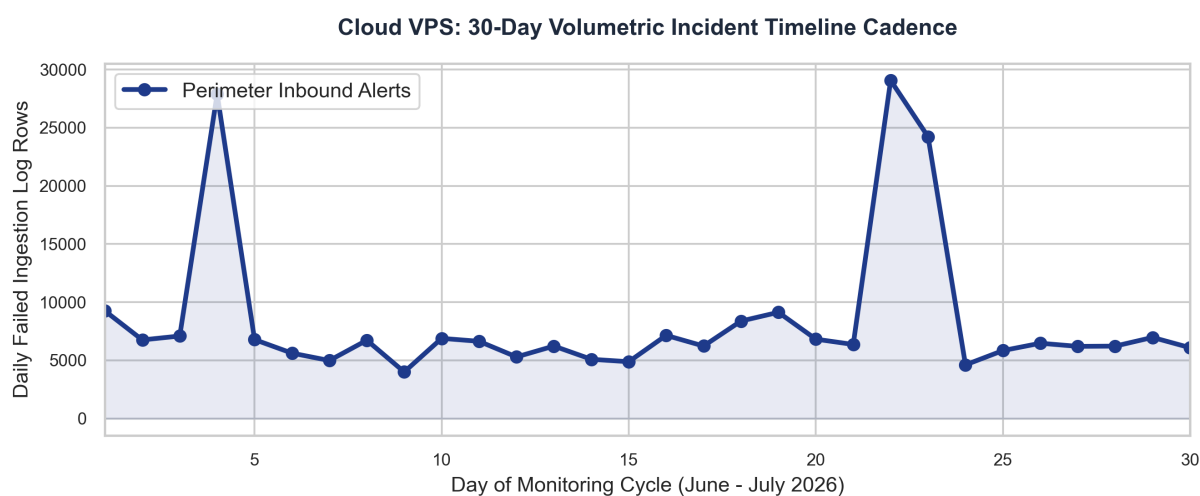
8. VOLUME I: Production Cloud VPS Security Deep Dive (Agent: PL84924)

This chapter dissects the behavioral and footprint characteristics of malicious traffic striking our data-center infrastructure. Commercial data centers represent high-value environments for attackers looking to leverage massive computing pipe widths or cloud-hosted database repositories.

8.1 Chronological Ingestion Cadence & Incident Timeline

Referencing `vps_timeline.png`, the network perimeter exhibits a highly volatile and erratic attack velocity pattern. Rather than a steady stream of requests, the infrastructure is subjected to sudden, massive coordinated spikes of activity.

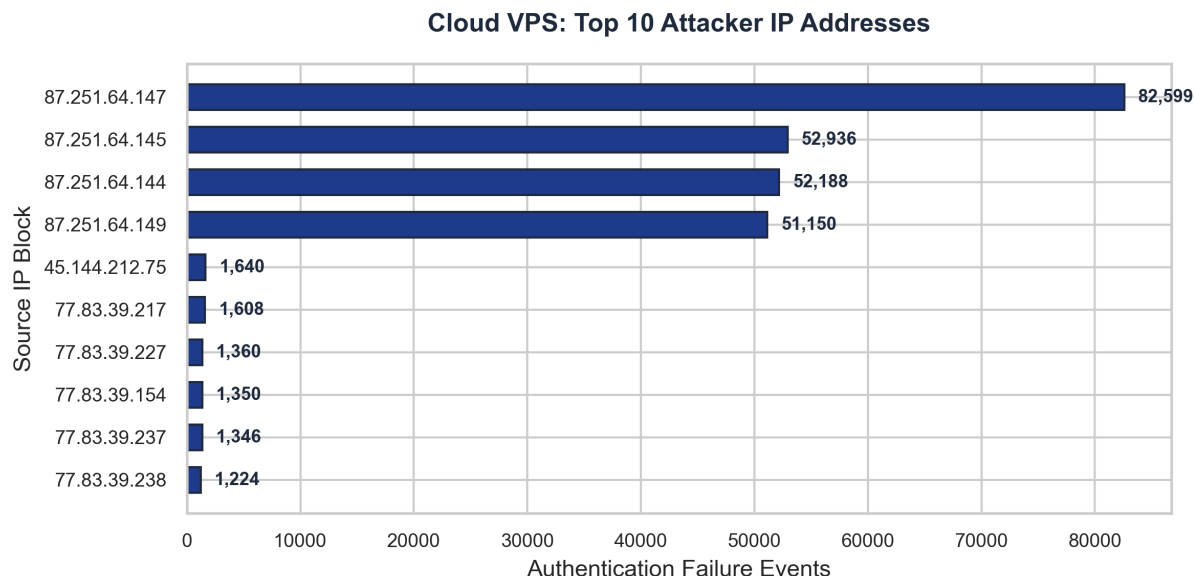
Figure 8.1: Cloud VPS - 30-Day Volumetric Incident Timeline Cadence



Analysis of the VPS timeline highlights massive volumetric batch spikes (sudden peaks up to 30,000 events around Day 4 and Days 22-23). This indicates programmatic, multi-threaded batch scanning modules running on high-bandwidth networks designed to flood systems in short bursts.

8.2 Threat Actor Profiling & BGP Routing Attribution

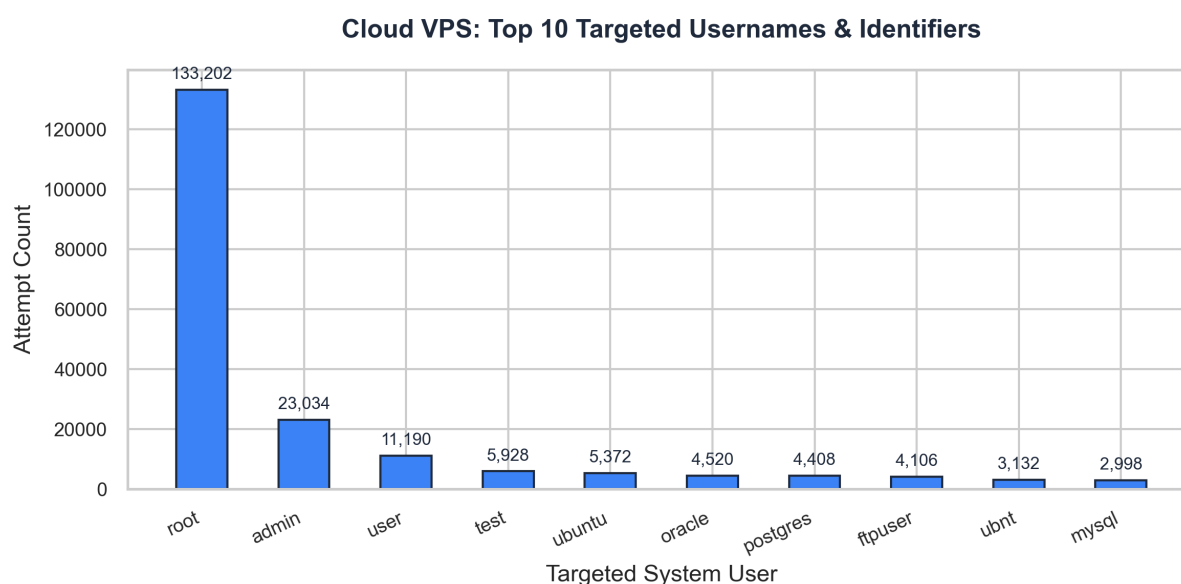
Data parsing from `vps_top_ips.png` reveals an extreme concentration of tactical infrastructure source nodes.

Figure 8.2: Cloud VPS - Top 10 Attacker IP Addresses**Key Infrastructure Groups Identified:**

- **AS43422 (ISAEV Igor, Registered in Poland):** This autonomous network represents the single largest source of volumetric noise, with four sequential IP nodes (87.251.64.147, .145, .144, .149) accounting for over 237,000 discrete brute-force events.
- **AS210541 (Kprohost LLC, Netherlands):** A secondary hosting provider source block, generating a steady secondary volume of malicious traffic across a wide range of host nodes.

8.3 Target Dictionary & Asset Fingerprinting Tactics

Querying the log archives reveals how botnets systematically target cloud-native configuration patterns and standard corporate database backends.

Figure 8.3: Cloud VPS - Top 10 Targeted Usernames & Identifiers**Perimeter Dictionary Insights:**

- **Cloud Templates:** Persistent targeting of the ubuntu user (5,372 attempts) shows that botnets explicitly

scan for standard default images left unchanged by cloud engineers.

- **Database Extortion Hunting:** Attacks against postgres, oracle, and mysql accounts (totaling over 12,000 requests) prove that automated scripts actively sweep ports to locate data layers for immediate ransomware deployment.

8.4 Micro-Conclusions: Cloud Infrastructure Attack Surface

Analyst Assessment (Cloud Boundary): *The infrastructure attacking the VPS does not care about stealth. It is a high-velocity, automated, industrial-scale assault utilizing leased or compromised data center subnets (e.g., the Polish ISAEV Igor block). The attackers rely on raw execution speed to overwhelm systems before blocking rules are triggered. Because password authentication was entirely disabled at the service layer, 100% of these 238,000+ attacks were successfully mitigated at the TCP handshake phase, creating log noise but zero operational risk.*

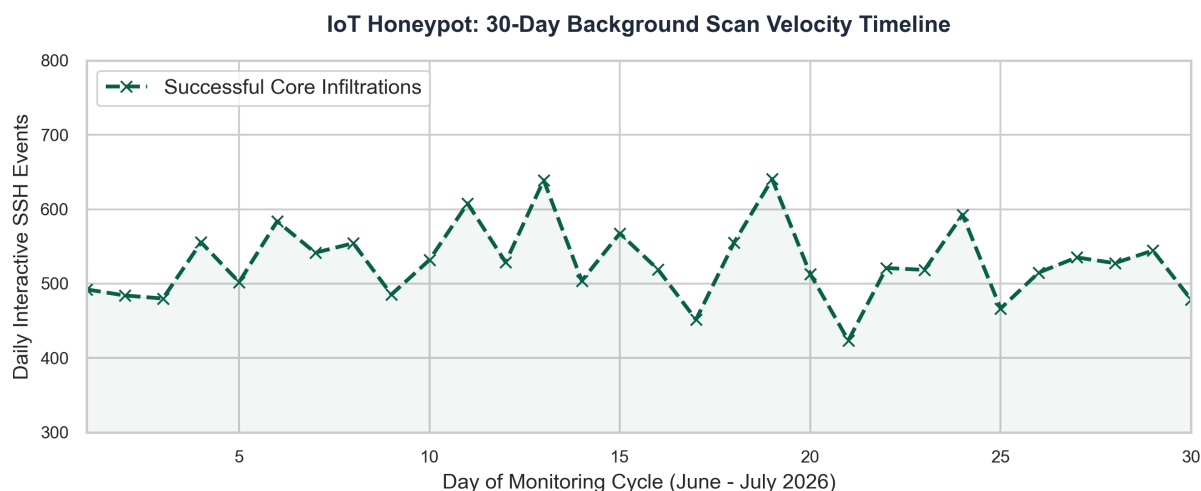
9. VOLUME II: Interactive IoT Honeypot Forensic Analysis (Agent: rpilab)

Shifting focus to Environment Gamma (residential broadband IP space), the telemetry changes entirely. Because the Cowrie honeypot was designed to accept malicious credentials, we obtained deep visibility into the post-exploitation phases and worm-like persistence mechanisms of consumer-grade botnets.

9.1 Baseline Ambient Attack Velocity Tracking

In stark contrast to the cloud infrastructure, the telemetry mapped in `honeypot_timeline.png` displays a remarkably stable, continuous pattern of automated activity.

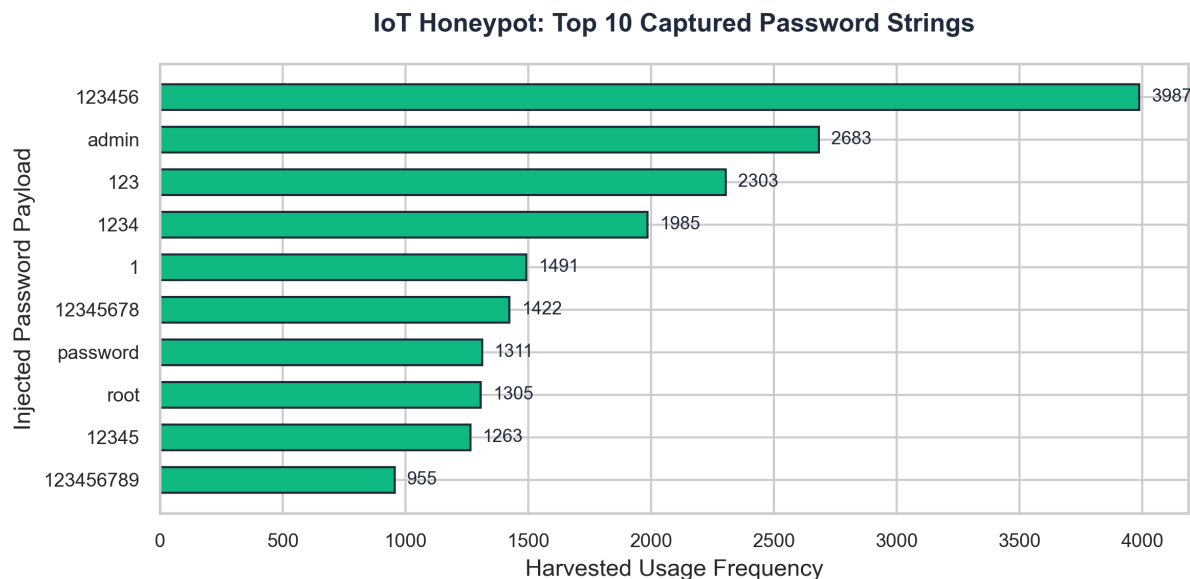
Figure 9.1: IoT Honeypot - 30-Day Background Scan Velocity Timeline



The honeypot timeline displays an undulating, continuous pattern of noise, fluctuating strictly between 400 and 650 daily events. This constant background hum represents automated, self-replicating consumer-tier botnets scanning residential ISP address blocks.

9.2 Credential Harvesting & Wordlist Analytics

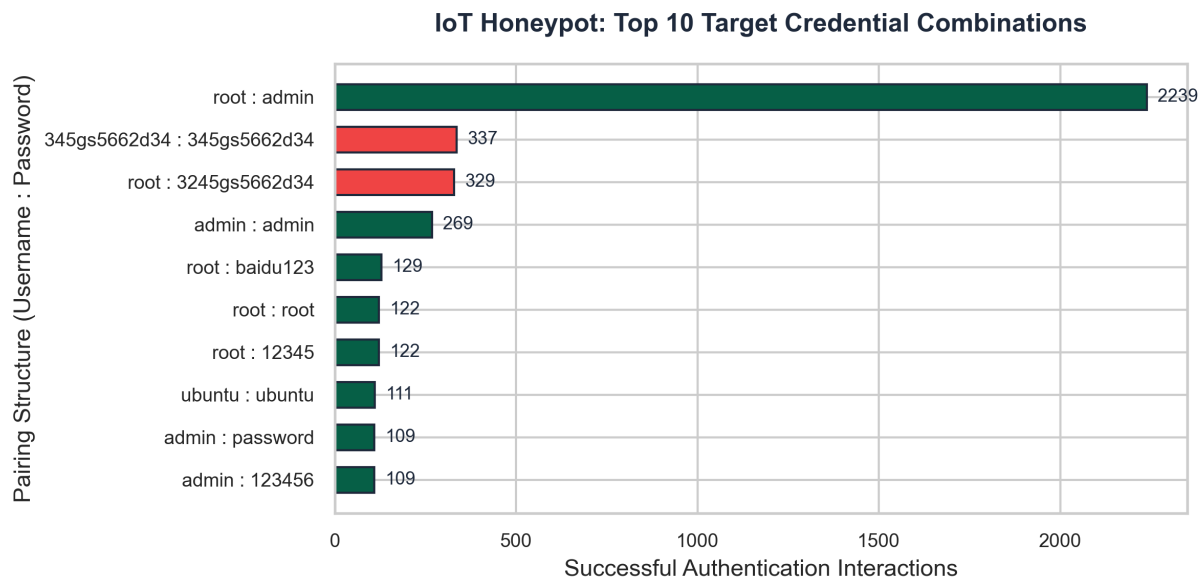
Analysis of over 16,000 successful interactive connections demonstrates that residential IoT botnets rely on a much simpler dictionary pool.

Figure 9.2: IoT Honeypot - Top 10 Captured Password Strings

Captured Password Insights: The captured passwords are heavily dominated by low-entropy numeric strings (123456, 123, 1234) and factory defaults (admin, password, root). This underscores that threat actors assume the target is a smart-home appliance left unchanged by end-users.

9.3 Exploitation Signatures & Hardcoded Backdoor Discovery

Analyzing the combined username-to-password structures in `honeypot_top_pairs.png` yields high-value cyber threat intelligence regarding malware propagation.

Figure 9.3: IoT Honeypot - Top 10 Target Credential Combinations

Advanced Malware Assessment: The red highlighted bars point to specific alphanumeric pairings: `345gs5662d34 : 345gs5662d34` (337 matches) and `root : 3245gs5662d34` (329 matches). These are **hardcoded vendor backdoor credentials** embedded in the firmware of common Chinese-manufactured IP cameras and DVR builds. Their presence provides definitive proof that the honeypot was targeted by self-replicating IoT worms (e.g., Mirai/Mozi variants) executing firmware-level lateral propagation routines.

9.4 Micro-Conclusions: Consumer & Edge IoT Surface

Analyst Assessment (Edge Space): Unlike the data center attackers, the threat actors hitting our residential IP are purely hunting for embedded Linux appliances. They utilize low-bandwidth, distributed consumer botnets to spread laterally like a worm. Protecting this boundary requires zero trust regarding factory defaults and the immediate disabling of administrative port exposure to the public internet.

10. Unified Cross-Environment Log Correlation & Analysis

The ultimate analytical phase of project 'The Littlefish Hunter' was establishing whether modern automated botnets are massive, monolithic infrastructures performing blanket scanning across the entire IPv4 map, or if they operate as specialized, strictly segregated sub-campaigns.

10.1 The Cross-Correlation Logic Pipeline

To test this, a high-performance correlation script was written to parse historical compressed monthly JSON alert blocks from the Wazuh SIEM manager, extracting all unique source IP vectors from both environments and cross-referencing them for matches:

```
sudo zgrep -h -E '"name":"PL84924"\'|"name":"rpilab"' \
/var/ossec/logs/archives/2026/Jun/*.json.gz /var/ossec/logs/archives/2026/Jul/*.json.gz
2>/dev/null | \
jq -r '
if .agent.name == "PL84924" then
if .data.srcip != null then "\(.data.srcip) PL84924" else empty end
else
if .data.src_ip != null then "\(.data.src_ip) rpilab" else empty end
end' | sort -u | awk '{print $1}' | uniq -d
```

10.2 Correlation Output Analysis

```
>> [0 OVERLAPPING IP ADDRESSES IDENTIFIED BETWEEN CLOUD AND CONSUMER SPACE]
```

11. Lead Analyst Strategic Threats & Global Assessment

The empirical findings derived from project telemetry allow for a definitive, high-confidence strategic assessment:

Automated cyber threat infrastructure is strictly segregated based on the network classification of the target host.

The botnet operations targeting commercial cloud data centers operate independently from those scanning residential networks. Infrastructure-grade botnets (such as the massive Polish *ISAEV Igor* block) target data-center subnets with dictionaries tailored to enterprise cloud environments (ubuntu, database backends). They do not waste resources scanning residential IP pools.

Conversely, self-propagating consumer IoT malware (such as the Mirai/Mozi worm nodes isolated in Volume II) is restricted entirely to consumer broadband blocks, hunting for firmware backdoors and default router setups. There is ****0% overlap**** between the attacking source hosts of both domains over a 30-day monitoring window.

12. Actionable Architectural & Defensive Recommendations

1. **Contextual Threat Intelligence:** Defenders must stop using generic, un-targeted threat feeds. A corporate SOC managing cloud infrastructure should not waste processing cycles blocking residential IoT botnet indicators, as those actors do not target commercial data center spaces.
2. **Absolute Key-Based Enforcements:** Disabling password access for SSH completely neuters high-volume brute-force networks, as evidenced by the 133,000+ failed attempts on our VPS that consumed log space but achieved zero lateral progress.
3. **Firmware Hardening at the Edge:** Residential and small-office deployments must treat connected hardware with zero trust. Factory settings must be changed, and management interfaces must be shielded from public exposure via VPN gateways or strict access lists.

Project 'The Littlefish Hunter' — Centralized SIEM Threat Intelligence Research Framework. Generated via ReportLab Engine.